3. Requirements Gathering

1. Stakeholder Analysis

Stakeholder	Role / Interest	Needs and Expectations	Influence
Job seekers	Understand AI-related changes to occupations and skills.	Clear job and industry trends; evidence-based automation-risk, opportunity, and skill insights.	High
Employers and HR teams	Assess AI-related role changes and workforce skill needs.	Comparable job and industry trends; defined indicators; useful workforce insights.	High
Policy makers and education providers	Inform workforce development, education, and reskilling decisions.	Aggregated trends, clear source context, and stated limitations.	Medium–high
Project team / data analysts	Prepare data, build the model and DAX measures, and develop the dashboard.	Defined scope, reliable data, testable measures, and dashboard validation.	High
Dataset administrator / data provider	Provides or maintains the analysis dataset.	Documented source, fields, date coverage, and known quality issues.	Medium–high
Lecturer / evaluator	Evaluates analysis, dashboard, testing, documentation, and presentation.	Reproducible analysis, explained measures, suitable visuals, and evidence-based conclusions.	High for academic evaluation

2. User Stories and Use Cases

2.1 User Stories

User Story	Expected Value
As a job seeker, I want to view how AI may affect different job categories so that I can understand	Relevant job-market trends are presented clearly, with the data period and limitations

potential changes in my field.	visible where available.
As an employer or HR user, I want to compare AI exposure and employment trends across industries and occupations so that I can identify areas of change.	Users can compare only categories supported by the dataset, using consistent definitions and measures.
As an education or workforce-planning user, I want to identify skills that are increasing in demand so that I can plan relevant training or reskilling.	Skills in demand are summarized from available data and are not presented as recommendations unsupported by the analysis.
As a dashboard user, I want to filter results by available industry, job category, skill, and time period so that I can focus on the information relevant to me.	Each available filter updates the relevant visuals and does not produce inconsistent totals.
As a project analyst, I want to validate the dataset and calculations so that dashboard findings are consistent and trustworthy.	Data-cleaning steps, model relationships, and DAX measures can be checked against documented test cases.

2.2 Use Cases

01 — Explore the impact of AI on jobs

Primary actor	Job seeker, employer/HR user, policy or education user
Preconditions	The approved dataset has been prepared, the data model is loaded, and the dashboard is available.
Main flow	1. The user opens the Power BI dashboard. 2. The dashboard displays key indicators and summary visuals. 3. The user reviews job displacement/automation risk, emerging opportunities, and skills trends as supported by the data. 4. The user selects a visual or category for more detail.
Alternative / exception	If a metric or field is unavailable, the dashboard omits it or clearly identifies it as unavailable. If no data matches a selection, the dashboard displays an empty/no-results state rather than a misleading value.
Postcondition	The user has viewed a data-supported summary and can distinguish observed information from interpretation.

02 — Filter and compare labor-market trends

Primary actor	Dashboard user

Preconditions	The dashboard contains supported slicers or filters for the selected data dimensions.
Main flow	1. The user selects an available industry, job category, skill, or time period. 2. The dashboard updates connected charts and indicators. 3. The user compares the filtered result with other selections or resets filters.
Alternative / exception	If the selected combination has no records, the dashboard communicates that no results are available. Unsupported dimensions are not offered as filters.
Postcondition	All affected visuals reflect the active filters consistently.

03 — Review skills in demand

Primary actor	Job seeker, employer/HR user, education provider
Preconditions	The approved data includes skill information suitable for analysis.
Main flow	1. The user opens the skills analysis view. 2. The dashboard presents the available skill-demand indicators. 3. The user filters by supported categories or periods. 4. The user reviews the results and their data context.
Alternative / exception	If skill data is missing or not comparable across records, the dashboard states the limitation and does not rank or compare unsupported values.
Postcondition	The user sees skill trends supported by the dataset and can interpret them within the stated scope.

3. Functional Requirements

Priority key: **M** = Must (required); **S** = Should (included if supported by the dataset and project time). Exact fields and measures must be confirmed against the real dataset.

	Functional Requirement	Acceptance / Verification
	The project shall load the approved job-market dataset and record its source, coverage period, and available fields.	M The dataset source and schema are documented; a valid data file loads successfully; unsupported fields are not assumed.
	The project shall clean and prepare data by handling missing values, duplicates, inconsistent formats, and incorrect data types as appropriate.	M Documented data-quality checks run on the dataset, and the treatment of each identified issue is recorded.
	The project shall build a data model with suitable	M Relationships are documented and

relationships for the available job categories, industries, skills, and time periods.		validated; visuals and measures return consistent results when filtered.
The project shall define DAX measures and calculated columns required to analyze employment trends, AI exposure, and skill demand, where supported by the data.	M	Each measure has a documented definition and returns expected results for selected test records or totals.
The Power BI dashboard shall present the project's main analysis areas: job displacement or automation risk, emerging job opportunities, and changes in job requirements and skills.	M	Each analysis area supported by the dataset appears in an appropriate visual or is explicitly documented as unavailable.
The dashboard shall display suitable KPIs, charts, slicers, and filters to support interactive exploration.	M	All included visuals have clear titles and units; each slicer filters the intended visuals correctly.
The dashboard shall support comparison across available job categories, industries, skills, and time periods.	M	Comparisons use consistent calculations and only dimensions present in the approved dataset.
The dashboard shall communicate the context of displayed results, including definitions, time range, and data source where available.	S	A reviewer can identify what each KPI represents and the data period/source used to calculate it.
The project shall present only data-supported conclusions and recommendations.	M	Each insight traces to a visual or measure; do not claim unsupported causation.

4. Non-Functional Requirements

	Requirement	**Verification**
	Data transformations, relationships, and DAX calculations shall produce consistent results from the approved source data.	Compare selected calculations with independently checked sample records or totals and document the result.
	Missing, duplicate, inconsistent, or invalid values shall be handled using documented rules rather than silently ignored.	Review the cleaning log and test representative data-quality cases.
	The dashboard shall use clear labels,	Ask representative users to locate a trend

understandable visual encodings, consistent navigation, and concise explanations suitable for non-technical users.	and apply a filter; record whether they can do so without explanation.
Dashboard visuals and filters shall respond within an acceptable time for the approved dataset and available execution environment.	Test interactions using the final dataset and document the environment and observed response time. Any numeric target should be agreed based on dataset size.
Filtering, totals, and charts shall remain consistent; empty selections	Test common filter combinations, empty results, and missing-field scenarios.
The project shall use data only for the approved academic purpose and shall not expose personal or confidential information in the dashboard.	Review dataset permissions and dashboard content; remove or aggregate personal data unless its use is explicitly authorized.
Definitions, data limitations, and interpretation limits shall be visible so users do not mistake AI exposure or automation risk for a guaranteed job outcome.	Review dashboard labels and notes; confirm indicators are defined and caveats are stated.
The dashboard shall be usable in the Power BI environment selected for project demonstration and submission.	Open and test the final dashboard in the agreed Power BI Desktop or Power BI Service environment.
Cleaning steps, the data model, DAX measures, and visuals shall be documented for review and updates.	The team can locate and explain each transformation, measure, and visual.